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| Semester            | S.E. Semester IV – INFT |
| Division & Batch    | Div A Batch 3           |
| Subject             | Python Programming      |
| Laboratory Teacher: | Prof. Pallavi Kharat    |
| Laboratory          | CC3                     |

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|---------------------------------------|------------------|
| Student Name                          | Advay Tongaonkar |
| Roll Number                           | 24101A0073       |
| Grade and Subject Teacher's Signature |                  |

| Experiment Number | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |               |                  |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
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| Experiment Title  | <div><div> Expt 10 Format.docx</div><div> Expt 10 Format.docx</div></div> <b>Pandas</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |               |                  |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
| Problem Statement | <p>(i) Develop a Python program using the Pandas library to analyze the sales data of a retail store.</p> <p>The program should create a DataFrame using the following input details:</p> <table><tr><th>Product ID</th><th>Product Name</th><th>Category</th><th>Quantity Sold</th><th>Price (per unit)</th></tr><tr><td>101</td><td>Laptop</td><td>Electronics</td><td>10</td><td>60000</td></tr><tr><td>102</td><td>Mobile</td><td>Electronics</td><td>25</td><td>25000</td></tr><tr><td>103</td><td>Tablet</td><td>Electronics</td><td>15</td><td>30000</td></tr><tr><td>104</td><td>Headphones</td><td>Accessories</td><td>40</td><td>2000</td></tr><tr><td>105</td><td>Smartwatch</td><td>Accessories</td><td>20</td><td>5000</td></tr></table> <p>Using this data, perform the following operations:</p> <ul style="list-style-type: none"><li>a) Create a Pandas DataFrame with appropriate column names.</li><li>b) Calculate the total sales amount for each product.</li><li>c) Compute the overall total revenue and average revenue.</li><li>d) Identify the best-selling product based on total sales value.</li></ul> | Product ID  | Product Name  | Category         | Quantity Sold | Price (per unit) | 101 | Laptop | Electronics | 10 | 60000 | 102 | Mobile | Electronics | 25 | 25000 | 103 | Tablet | Electronics | 15 | 30000 | 104 | Headphones | Accessories | 40 | 2000 | 105 | Smartwatch | Accessories | 20 | 5000 |
| Product ID        | Product Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Category    | Quantity Sold | Price (per unit) |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
| 101               | Laptop                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Electronics | 10            | 60000            |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
| 102               | Mobile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Electronics | 25            | 25000            |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
| 103               | Tablet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Electronics | 15            | 30000            |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
| 104               | Headphones                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Accessories | 40            | 2000             |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |
| 105               | Smartwatch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Accessories | 20            | 5000             |               |                  |     |        |             |    |       |     |        |             |    |       |     |        |             |    |       |     |            |             |    |      |     |            |             |    |      |

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|                                | <p>e) Display products whose total sales exceed ₹5,00,000.</p> <p>Export to Excel</p> <p>Display category-wise total revenue</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
| Resources / Apparatus Required | Hardware: Desktop/Laptop                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Software: |
| Code                           | <pre> import pandas as pd  data = {     "Product ID": [101, 102, 103, 104, 105],     "Product Name": ["Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch"],     "Category": ["Electronics", "Electronics", "Electronics", "Accessories", "Accessories"],     "Quantity Sold": [10, 25, 15, 40, 20],     "Price per Unit": [60000, 25000, 30000, 2000, 5000] }  df = pd.DataFrame(data)  print("----- Product Data --- --") print(df)  df["Total Sales"] = df["Quantity Sold"] * df["Price per Unit"]  print("\n----- Data with Total Sales -----") print(df)  total_revenue = df["Total Sales"].sum()  average_revenue = df["Total Sales"].mean()  print("\nOverall Total Revenue = ₹", total_revenue) </pre> |           |

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|  | <pre>print("Average Revenue = ₹", average_revenue)  best_product = df.loc[df["Total Sales"].idxmax()]  print("\n----- Best Selling Product -----") print(best_product)  high_sales = df[df["Total Sales"] &gt; 500000]  print("\n----- Products with Sales &gt; ₹5,00,000 -----") print(high_sales)  category_revenue = df.groupby("Category")["To tal Sales"].sum()  print("\n----- Category-wise Total Revenue -----") print(category_revenue)</pre> |
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| <p>output</p>     | <div><div><div>main.py</div><div><pre>1 import pandas as pd 2 3 data = { 4     "Product ID": [101, 102, 103, 104, 105], 5     "Product Name": ["Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch"], 6     "Category": ["Electronics", "Electronics", "Electronics", "Accessories", "Accessories"], 7     "Quantity Sold": [10, 25, 15, 40, 20], 8     "Price per Unit": [60000, 25000, 30000, 2000, 5000] 9 } 10 11 df = pd.DataFrame(data) 12 13 print("----- Product Data -----") 14 print(df) 15 16 df["Total Sales"] = df["Quantity Sold"] * df["Price per Unit"] 17 18 print("\n----- Data with Total Sales -----") 19 print(df) 20 21 total_revenue = df["Total Sales"].sum() 22 23 average_revenue = df["Total Sales"].mean() 24 25 print(f"\nOverall Total Revenue = ₹ {total_revenue}") 26 print(f"Average Revenue = ₹ {average_revenue}") 27 28 best_product = df.loc[df["Total Sales"].idxmax()] 29 30 print("\n----- Best Selling Product -----") 31 print(best_product) 32 33 high_sales = df[df["Total Sales"] &gt; 500000] 34 35 print("\n----- Products with Sales &gt; ₹5,00,000 -----") 36 print(high_sales) 37 38 category_revenue = df.groupby("Category")["Total Sales"].sum() 39 40 print("\n----- Category-wise Total Revenue -----") 41 print(category_revenue)</pre></div><div>Output</div><div><pre>Product ID Product Name Category Quantity Sold Price per Unit 0 101 Laptop Electronics 10 60000 1 102 Mobile Electronics 25 25000 2 103 Tablet Electronics 15 30000 3 104 Headphones Accessories 40 2000 4 105 Smartwatch Accessories 20 5000  ----- Data with Total Sales ----- Product ID Product Name ... Price per Unit Total Sales 0 101 Laptop ... 60000 600000 1 102 Mobile ... 25000 625000 2 103 Tablet ... 30000 450000 3 104 Headphones ... 2000 80000 4 105 Smartwatch ... 5000 100000  [5 rows x 6 columns]  Overall Total Revenue = ₹ 1855000 Average Revenue = ₹ 371000.0  ----- Best Selling Product ----- Product ID    102 Product Name  Mobile Category      Electronics Quantity Sold    25 Price per Unit   25000 Total Sales    625000 Name: 1, dtype: object  ----- Products with Sales &gt; ₹5,00,000 ----- Product ID Product Name ... Price per Unit Total Sales 0 101 Laptop ... 60000 600000 1 102 Mobile ... 25000 625000  [2 rows x 6 columns]  ----- Category-wise Total Revenue ----- Category Accessories    180000 Electronics   1675000 Name: Total Sales, dtype: int64</pre></div></div></div> |
| <p>Conclusion</p> | <p>The program analyzes retail store sales data using Pandas by calculating total sales, revenue, and identifying high-performing products. It helps in understanding category-wise revenue distribution and finding the best-selling products based on total sales value.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |